AI-Driven Public Transport Image Classification for Road Safety and Efficiency

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Capstone Project: Group 1

Introduction

Background:

- Kenya's public transport system is vital for millions depending on matatus, buses, tuk-tuks, and boda bodas.

- The public transport sector generates over KSh 200 billion each year and provides direct employment to about 250,000 people

Beba Beba AI – Smart Transport Monitoring:

- Problem Statement: Vehicle monitoring in Kenya's public transport system causes inefficiencies, contributing to congestion and regulatory issues.

- The Solution: AI-powered image classification for real-time compliance monitoring and fleet management for safer roads and better traffic flow.

- How It Works: Uses AI to classify PSVs (matatus, buses, tuk-tuks, bodas) and detect unsafe or overloaded vehicles.

Data Understanding

- Sources: Web scraping (Google Images, OpenStreetMap), public datasets, manual labelling.(e.g., Fatkun Extension).
- Vehicle Categories: Matatus, Buses , Boda-Bodas, Tuk-Tuks - Approx 800 images collected.
- Additionally under Buses we also gathered images of various PSV Saccos e.g Super Metro, KBS, City Hopper etc.
- Quality Criteria: High resolution, proper lighting, clear perspectives, no obstructions.
- Images were mainly in jpg format.

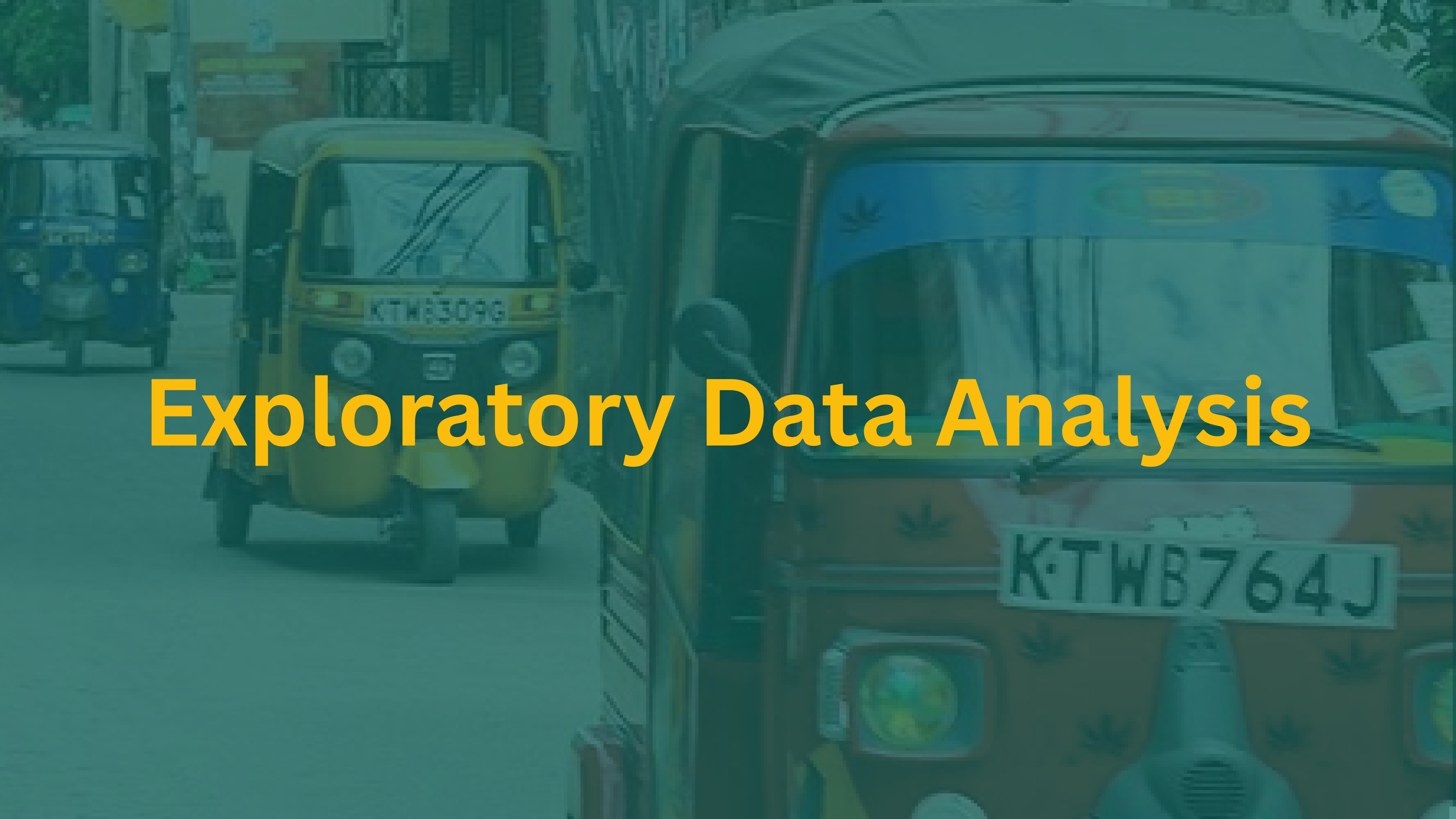
OBJECTIVES

- **Objective 1:** Develop an AI system to classify PSVs and assist law enforcement in detecting non-compliant vehicles to enhance road safety.

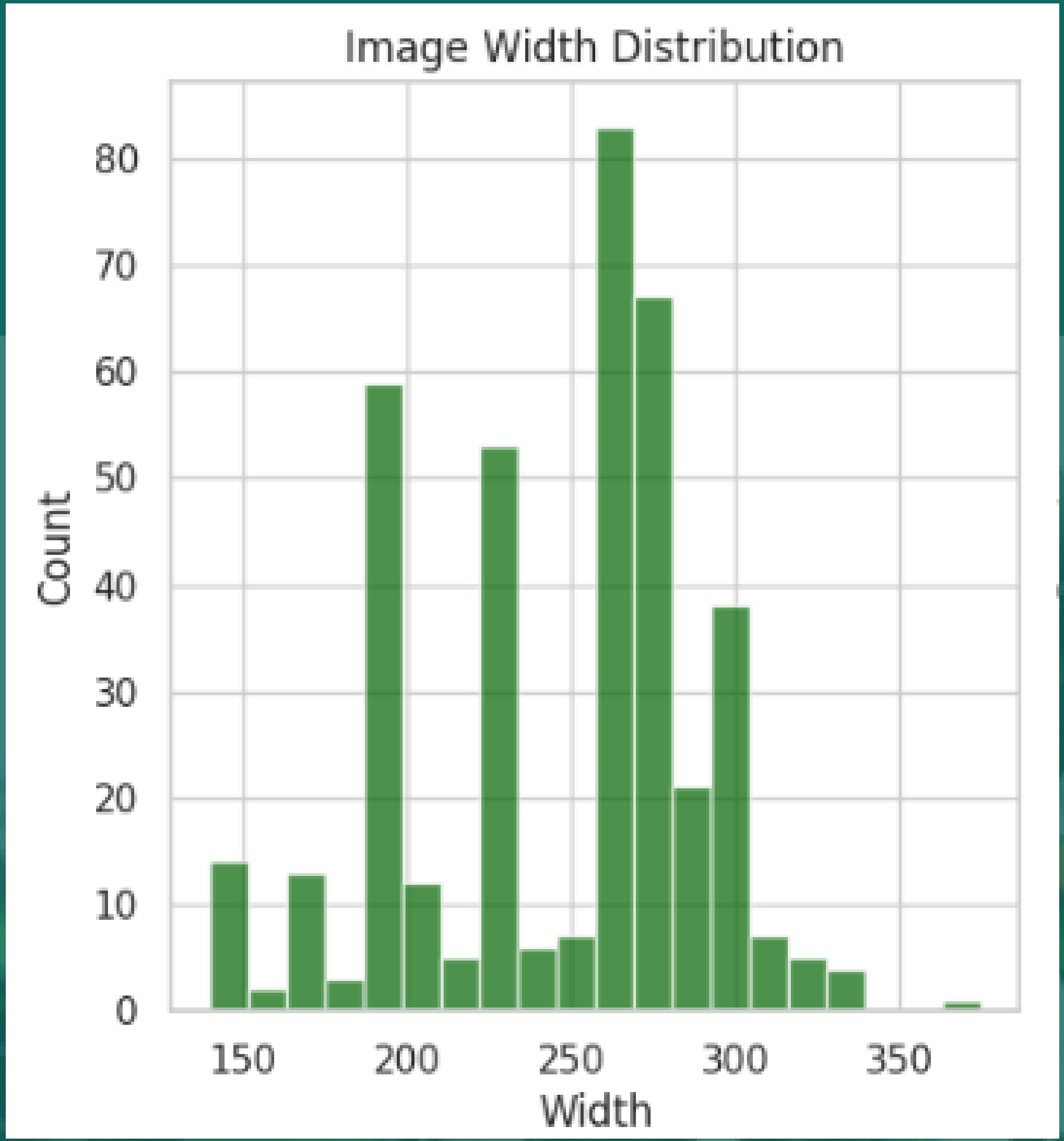
- **Objective 2:** To build a system that can identify and track Public Service Vehicles in real-time using YOLOv8, a deep-learning model.

- **Objective 3:** Automate fleet tracking and classification for transport operators to monitor vehicle conditions, optimize maintenance, and ensure regulatory compliance.

Exploratory Data Analysis



- Image Dimensions for PSV Categories: Varying sizes suggest resizing or cropping for consistency.



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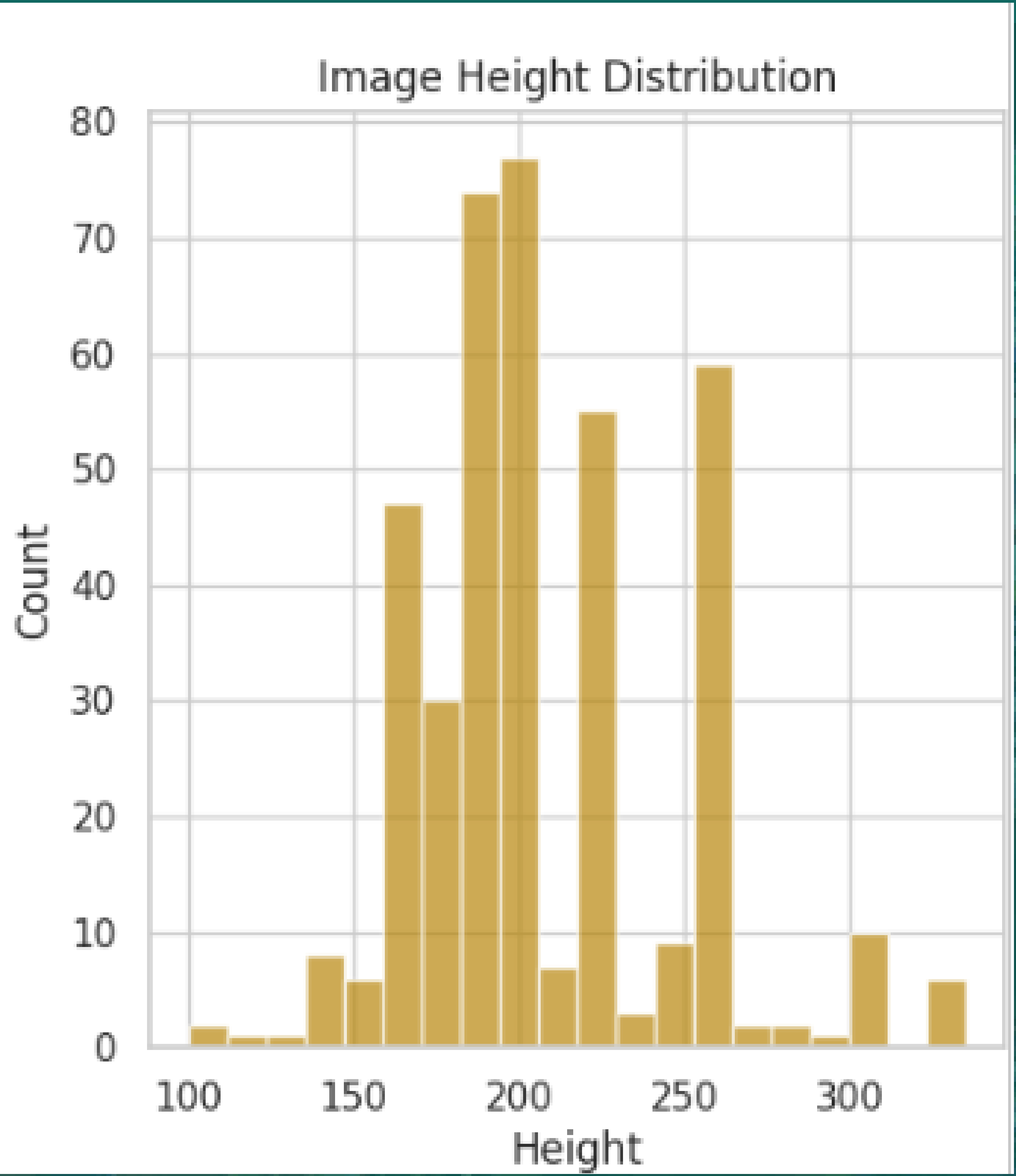
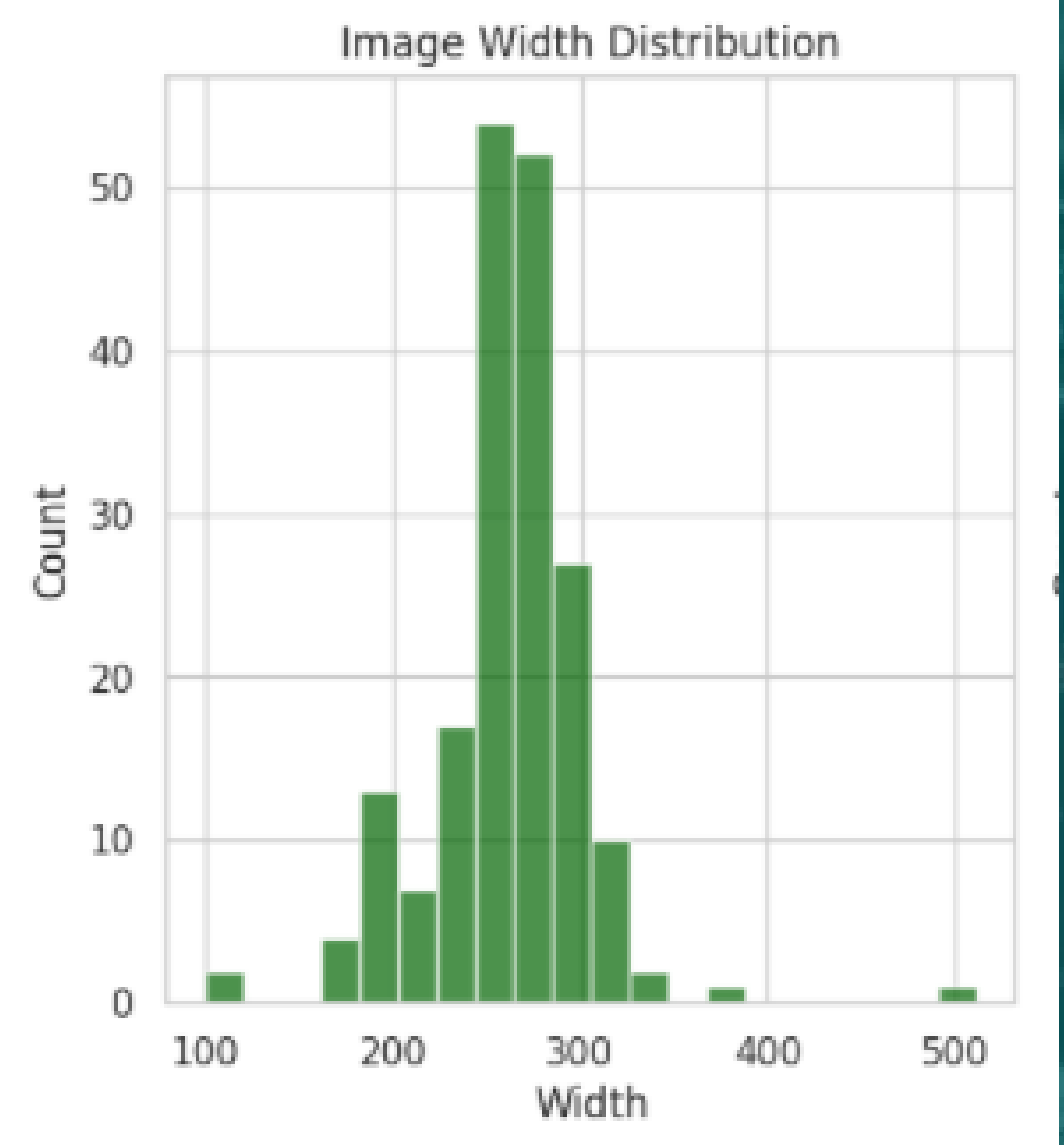
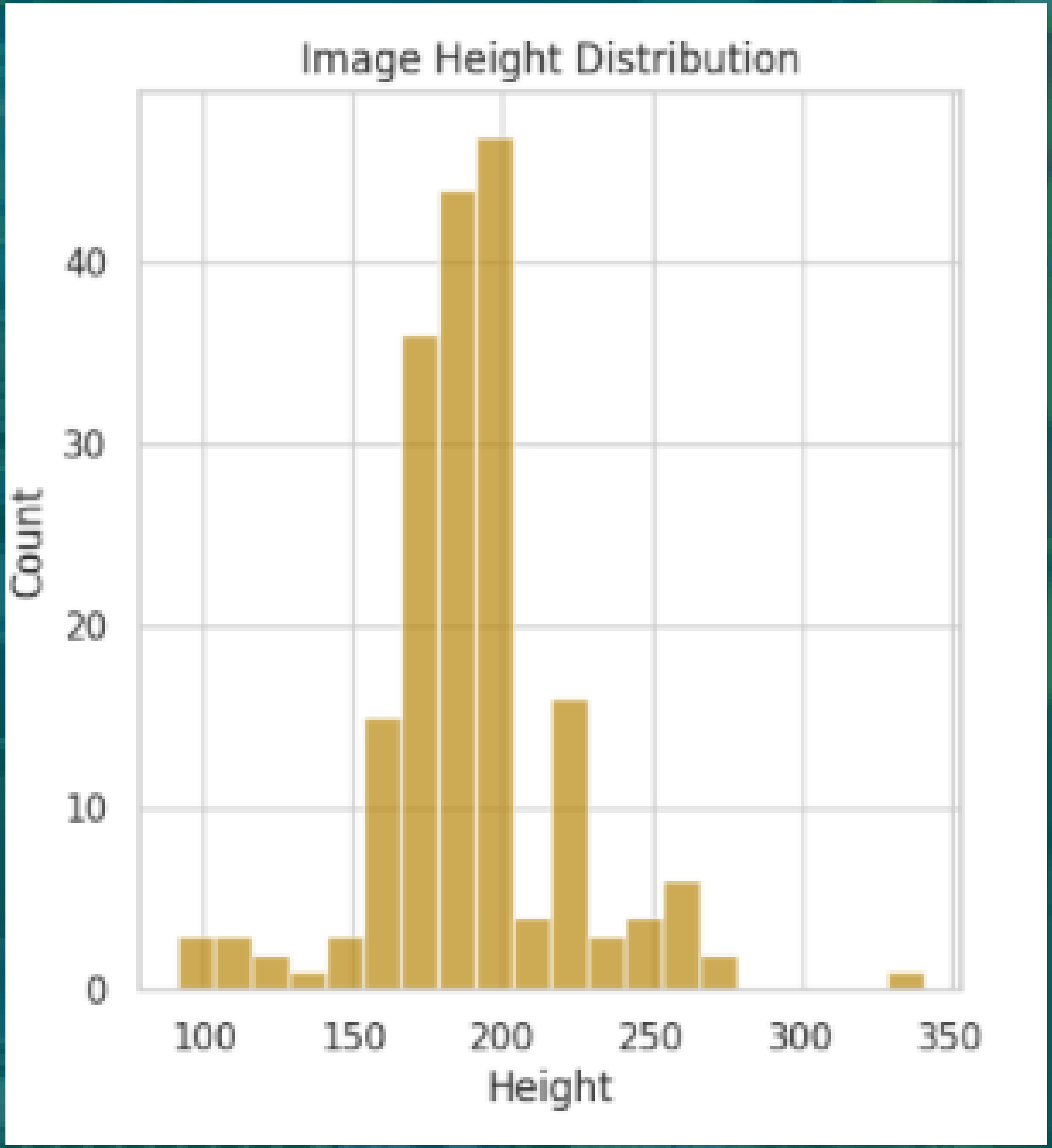


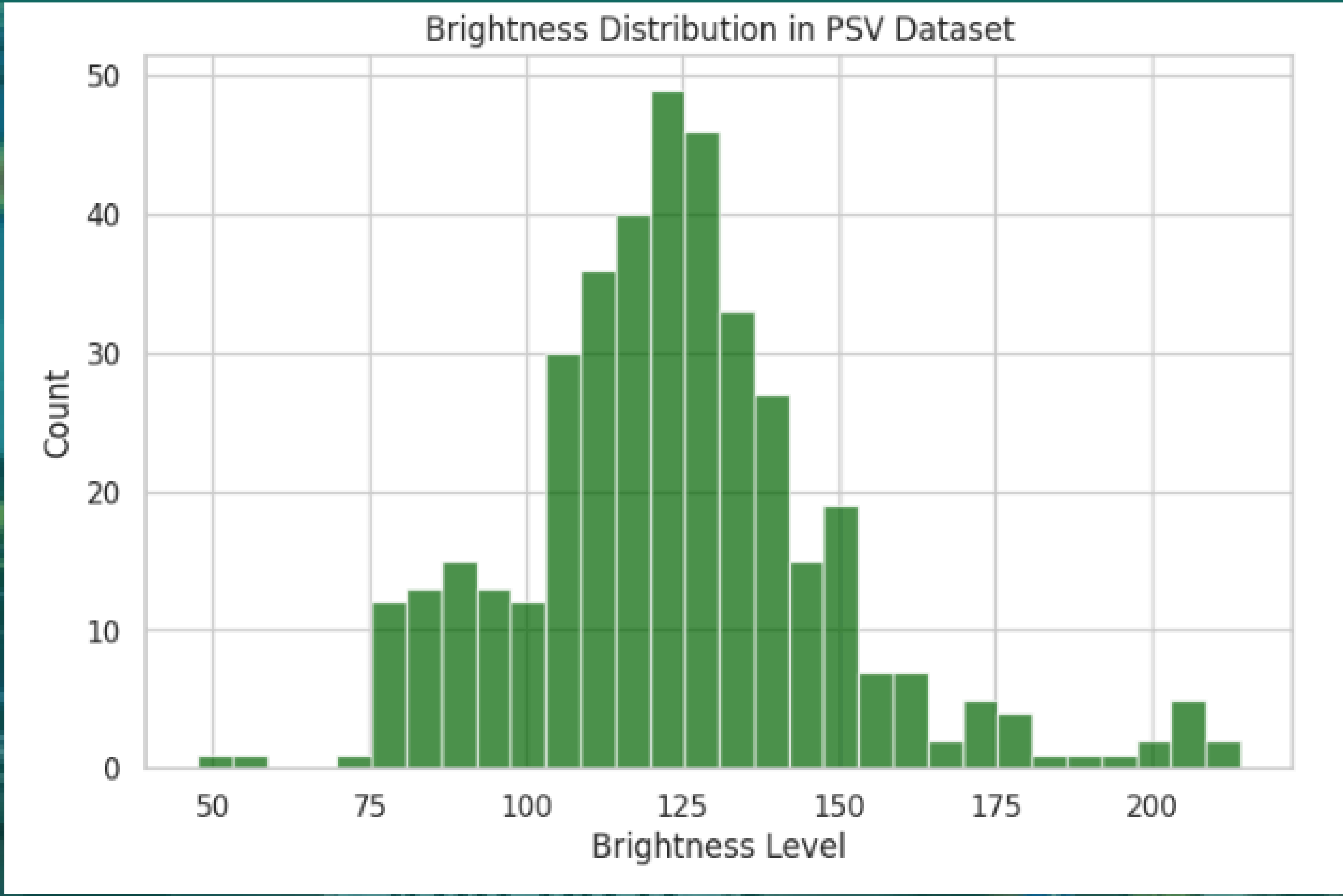
- Image Dimensions for Bus Categories: Varying sizes suggest resizing or cropping for consistency.



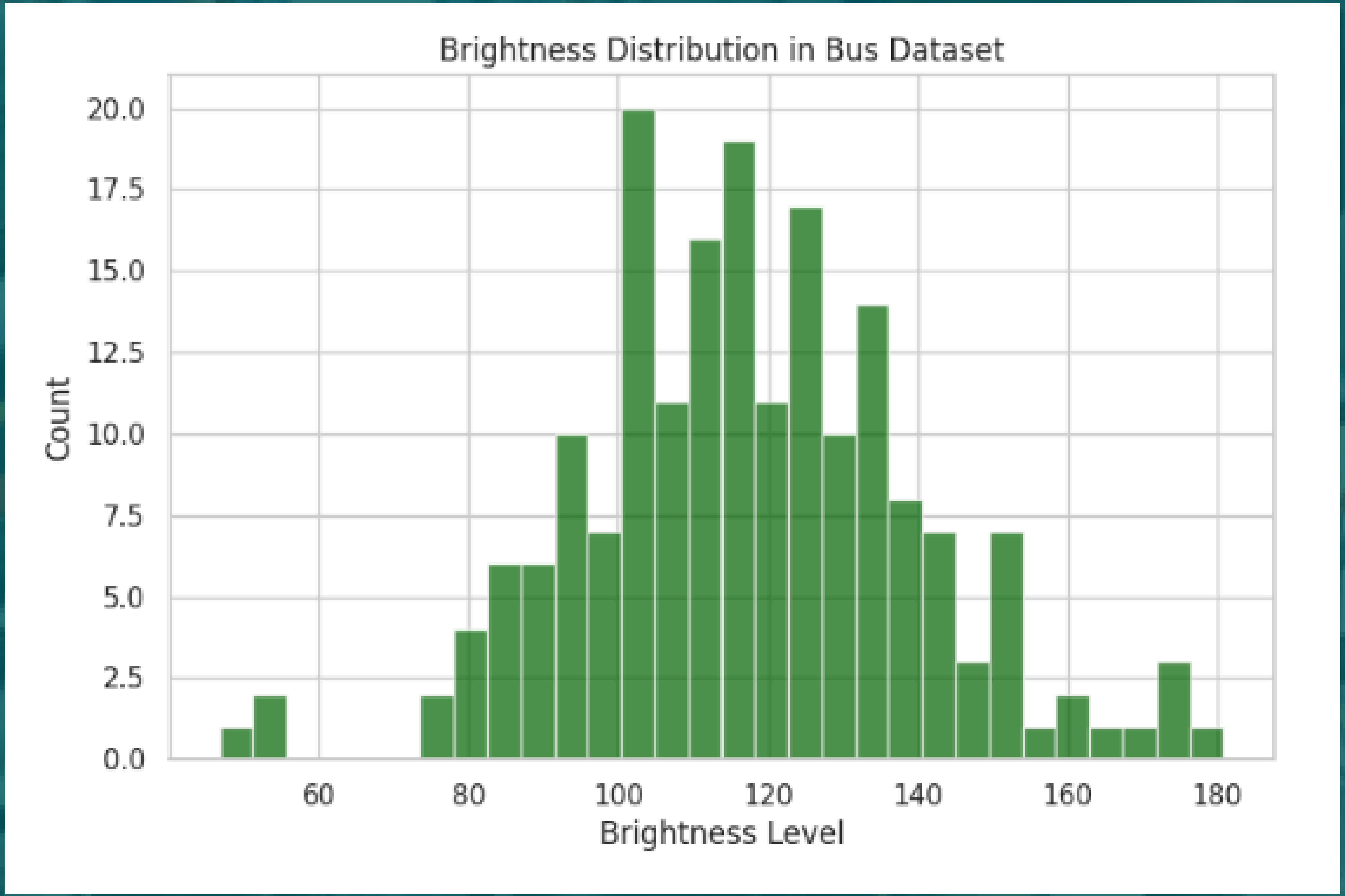
- **Image Dimensions for Bus Categories: Varying sizes suggest resizing or cropping for consistency.**



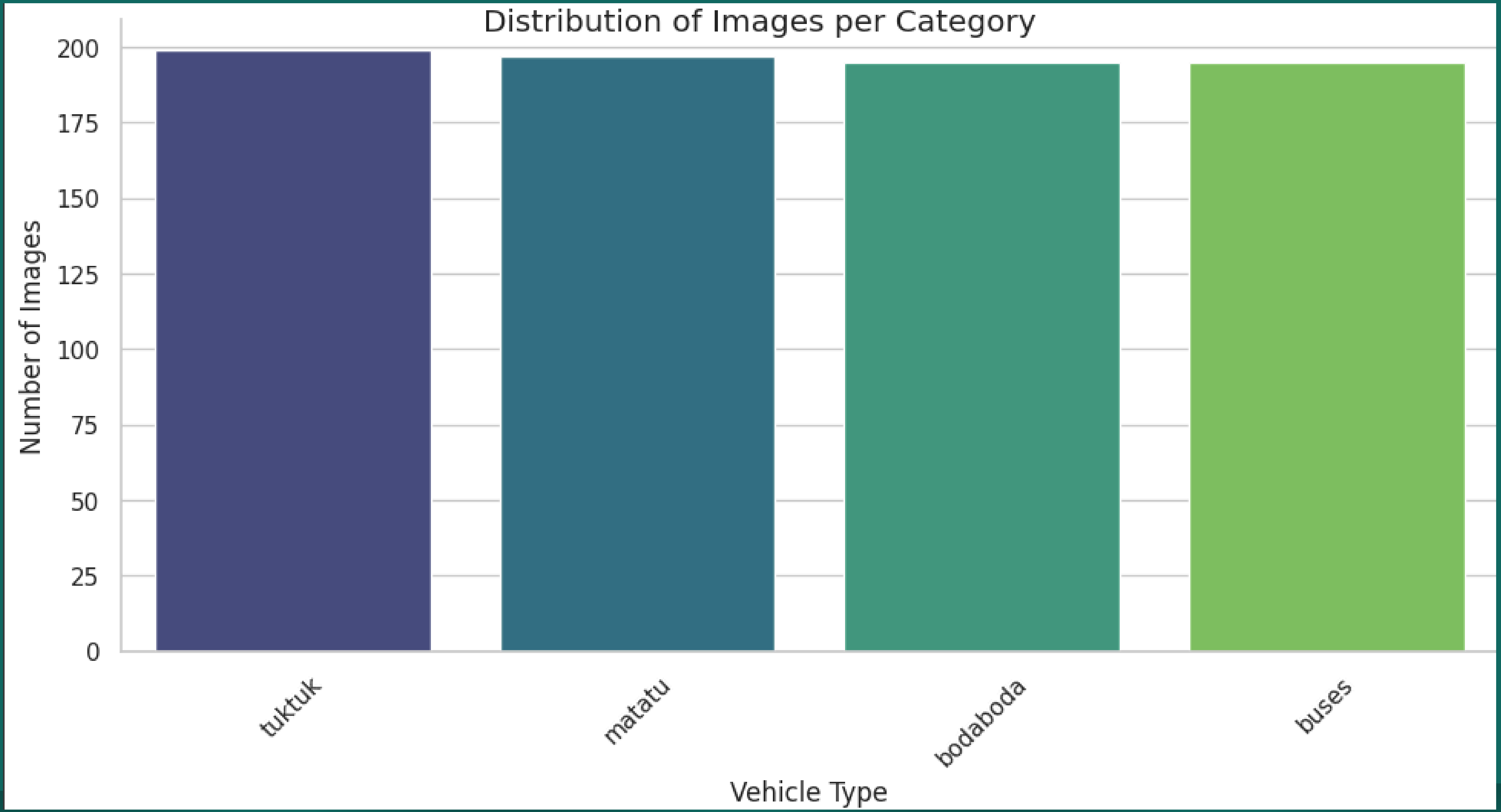
- Image Brightness for PSV Categories: Follows a normal distribution, ensuring uniform lighting for model training.



- **Brightness for Bus Categories: Follows a normal distribution, ensuring uniform lighting for model training.**

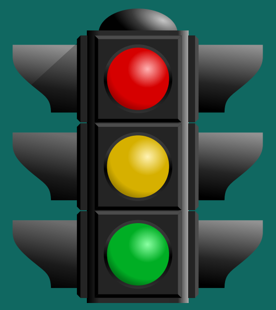


- Image count and distribution per category: Data was evenly distributed among the four categories

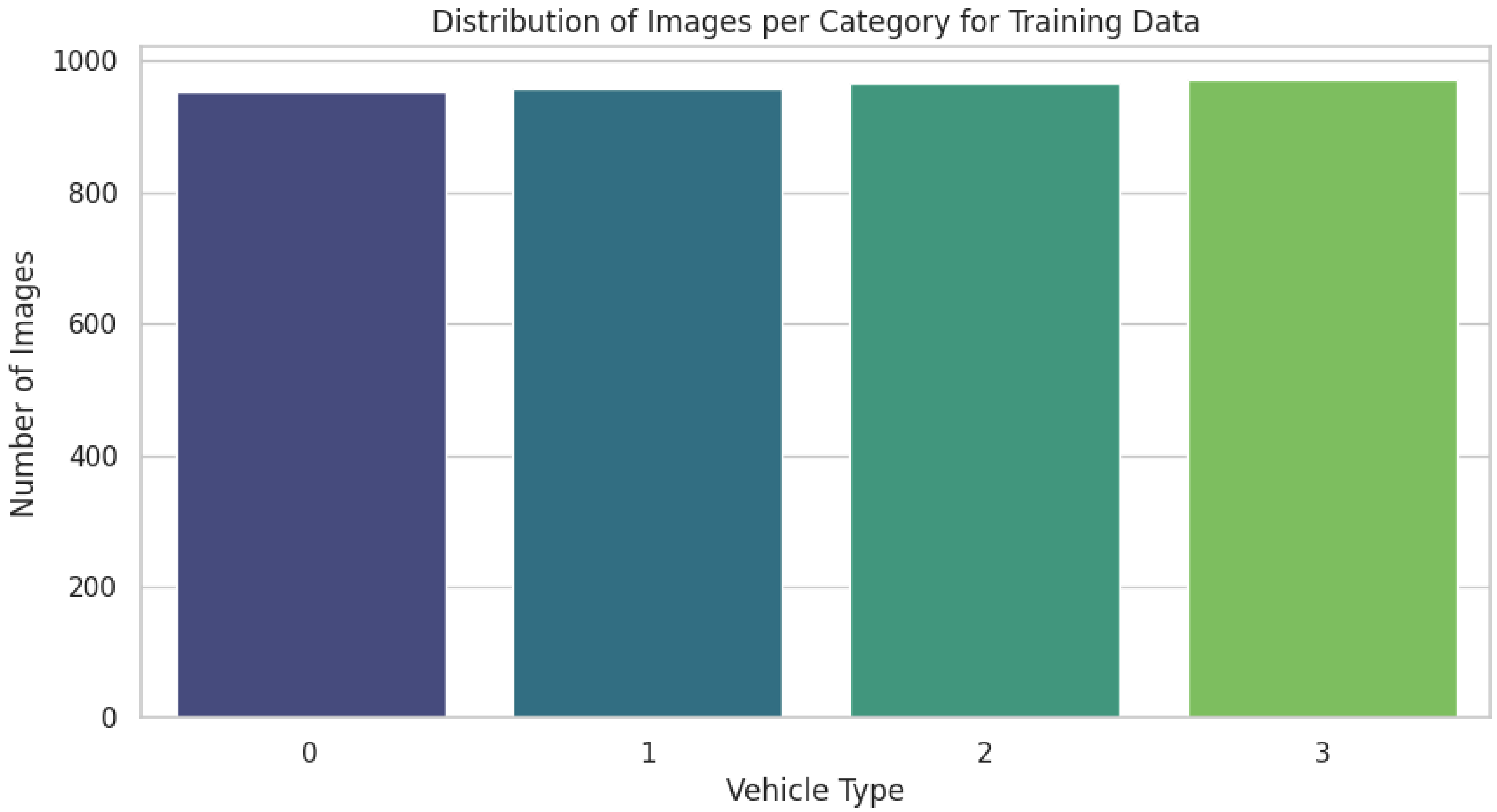


Data Preparation & Preprocessing:

- Manual data cleaning: Visually inspected the images and removed blurry images and duplicates using the Norton Duplicate File Finder.
- Criteria for images: Blurry, pixelated, distorted, overexposed or underexposed, and images with excessive distractions in the background.
- Additional validations: Check for corrupted images and ensure each image has a valid image extension (.jpg, .jpeg, .png, .gif, .bmp).
- Standardization: Renamed images for consistency, stored in a structured data frame.
- Scaling & Splitting: Normalized pixel values (0-1) and divided data (70% Train, 15% Validation, 15% Test).
- Augmentation & Label Encoding: Applied flips, blurs, brightness/contrast adjustments, distortions and rotations to enhance model robustness.
- The Roboflow app was used to label and annotate each image.

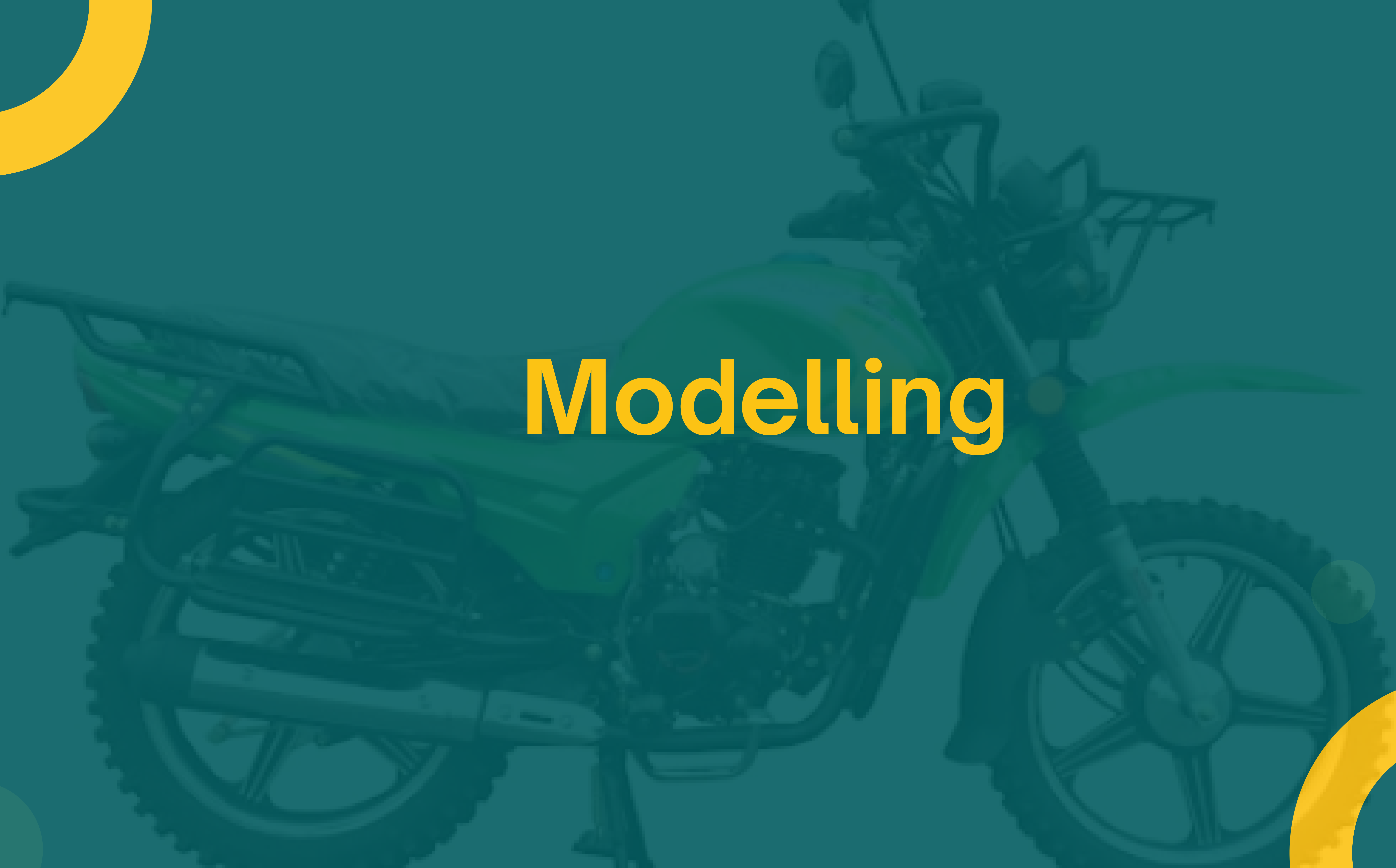


- **Post Augmentation distribution for PSVs**



- **Post Augmentation distribution for Buses**

Modelling



Objective 1: PSV image classification and compliance monitoring

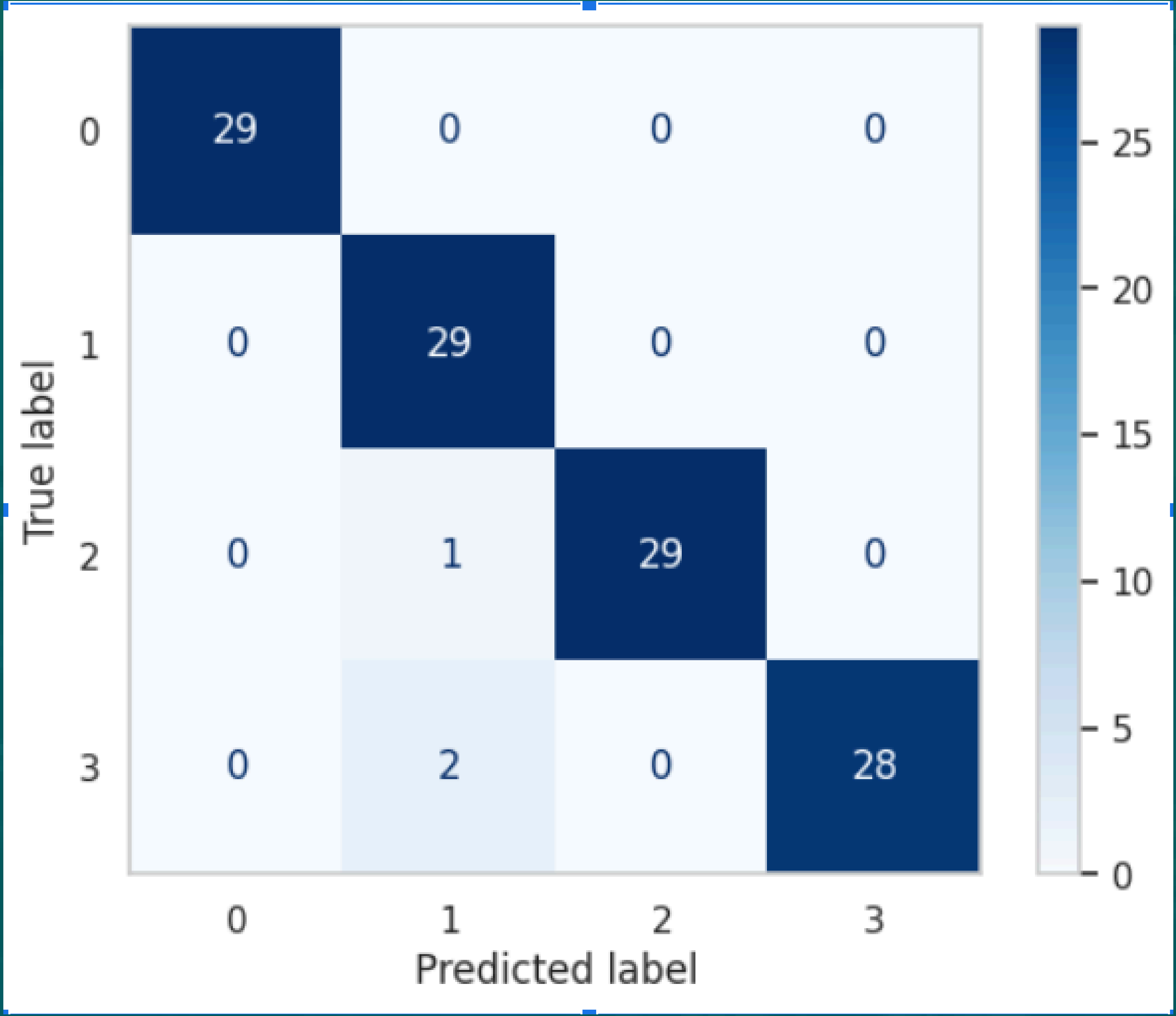
A two-stage model is used for vehicle overloading detection:

- Vehicle Classification Model – Identifies the type of vehicle (matatus, buses, tuk-tuks, boda-bodas).
- Overloading Detection Model – Determines if the classified vehicle is overloaded.

1.Vehicle Classification Model

- Baseline Convolutional Neural Network: 98.5% training accuracy, but 60% validation accuracy → Overfitting.
- Tuning (Dropout, Learning Rate, Augmentation): Improved generalization, but accuracy plateaued.
- Transfer Learning (MobileNetV2): Achieved ~99% training, 97-98% validation accuracy with minimal overfitting.

Confusion Matrix



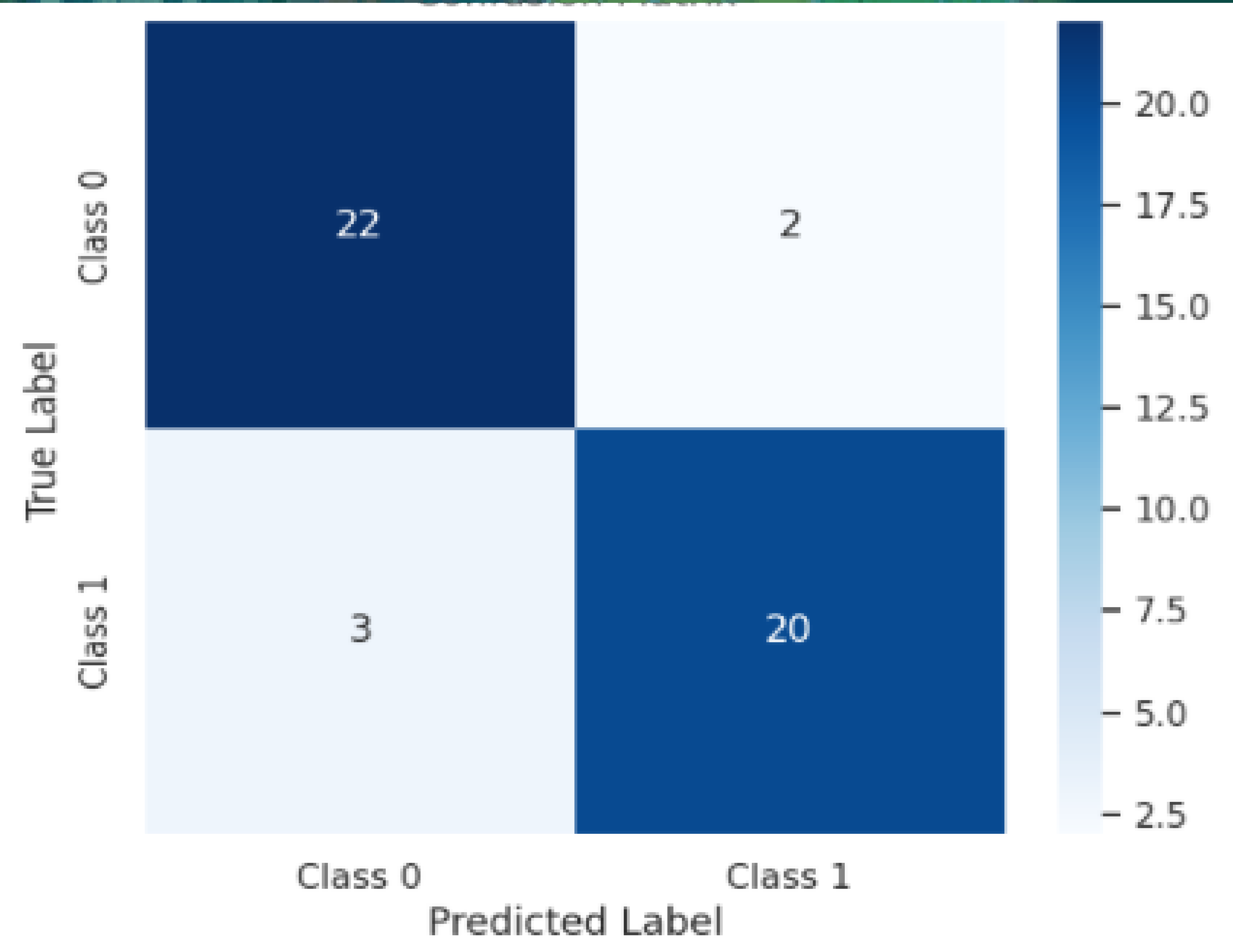
- **Class 0(Boda Boda):** 29 correct predictions.
- **Class 1(Bus):** 29 correct predictions, with 1 misclassified as Class 2.
- **Class 2(Matatu):** 29 correct predictions, with 1 misclassified as Class 1.
- **Class 3(Tuk Tuk):** 28 correct predictions, with 2 misclassified as Class 1.

Objective 1: PSV image classification and compliance monitoring

2. **Overloading Detection.**

- Initial Convolutional Neural Network overfitted (100% training accuracy, 70% validation).
- Transfer Learning (MobileNetV2) reduced overfitting with fine-tuning, regularization, and SGD optimizer.
- Final Model: 94% accuracy, balanced precision & recall, strong generalization.

Confusion Matrix



- 22 correct predictions for Class 0 (Not Overloaded).
- 20 correct predictions for Class 1 (Overloaded).
- 2 misclassifications where Class 0 was predicted as Class 1 (False Positives).
- 3 misclassifications where Class 1 was predicted as Class 0 (False Negatives).

Objective 2: Real-time PSV detection using YOLOv8

1.Baseline Model: YOLOv8s

Initial Performance Metrics:

- Precision: 85.1%, Recall: 73.6%,mAP50 (Accuracy at 50% threshold): 83.4%

Class-wise Performance:

The model faced challenges in capturing more bodabodas and buses.

- Bodaboda: Precision: 94.2%, Recall: 55.8%,mAP50: 78.7%

- Bus: Precision: 75.1%, Recall: 69%,mAP50: 80.6%

To address these challenges, recall needed improvement, prompting the need for further model tuning.

Objective 2: Real-time PSV detection using YOLOv8

2. YOLOv8m (More Complex Version)

To enhance detection, the model was upgraded to YOLOv8m, a more advanced version of YOLO designed for handling more complex tasks.

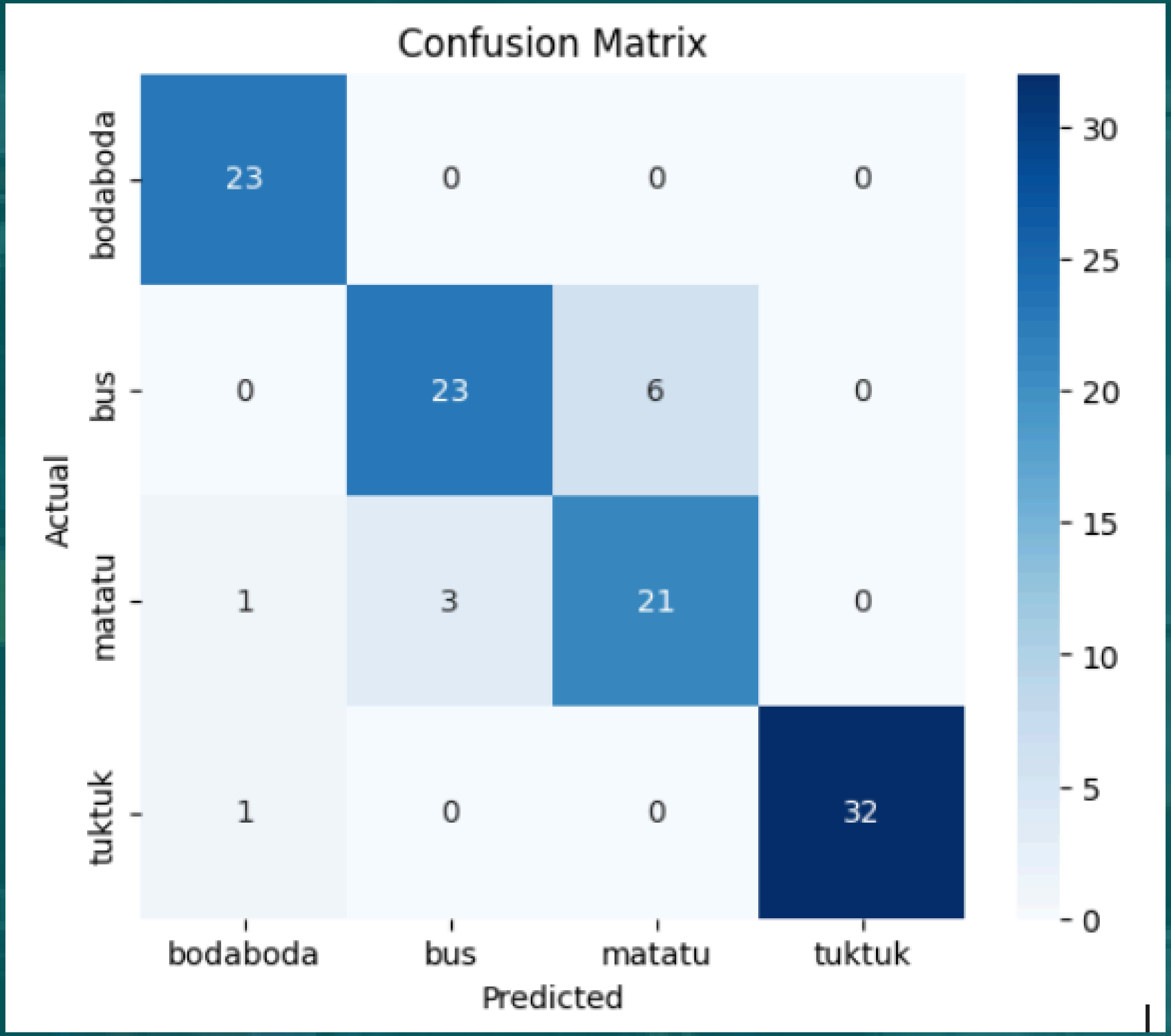
Performance with YOLOv8m:

- Precision: 85.9%
- Recall: 79%
- mAP50 (Accuracy at 50% threshold): 86.5%

The model showed improved performance, particularly with a higher recall rate, especially for detecting bodabodas and buses.

Class-wise Performance with YOLOv8m:

- For bodabodas and buses, recall improved, which is a positive sign.
  - Bodaboda: Precision: 90.5%, Recall: 65.5%
  - Bus: Precision: 71.9%, Recall: 75.9%



Class 0 (bodaboda): 23 correct predictions, 2 misclassified (1 as Class 2, 1 as Class 3).
Class 1 (bus): 23 correct predictions, 6 misclassified as Class 2.
Class 2 (matatu): 21 correct predictions, 3 misclassified as Class 1, 1 as Class 0.
Class 3 (tuktuk): 32 correct predictions, 1 misclassified as Class 0.

Objective 3: Bus fleet image classification

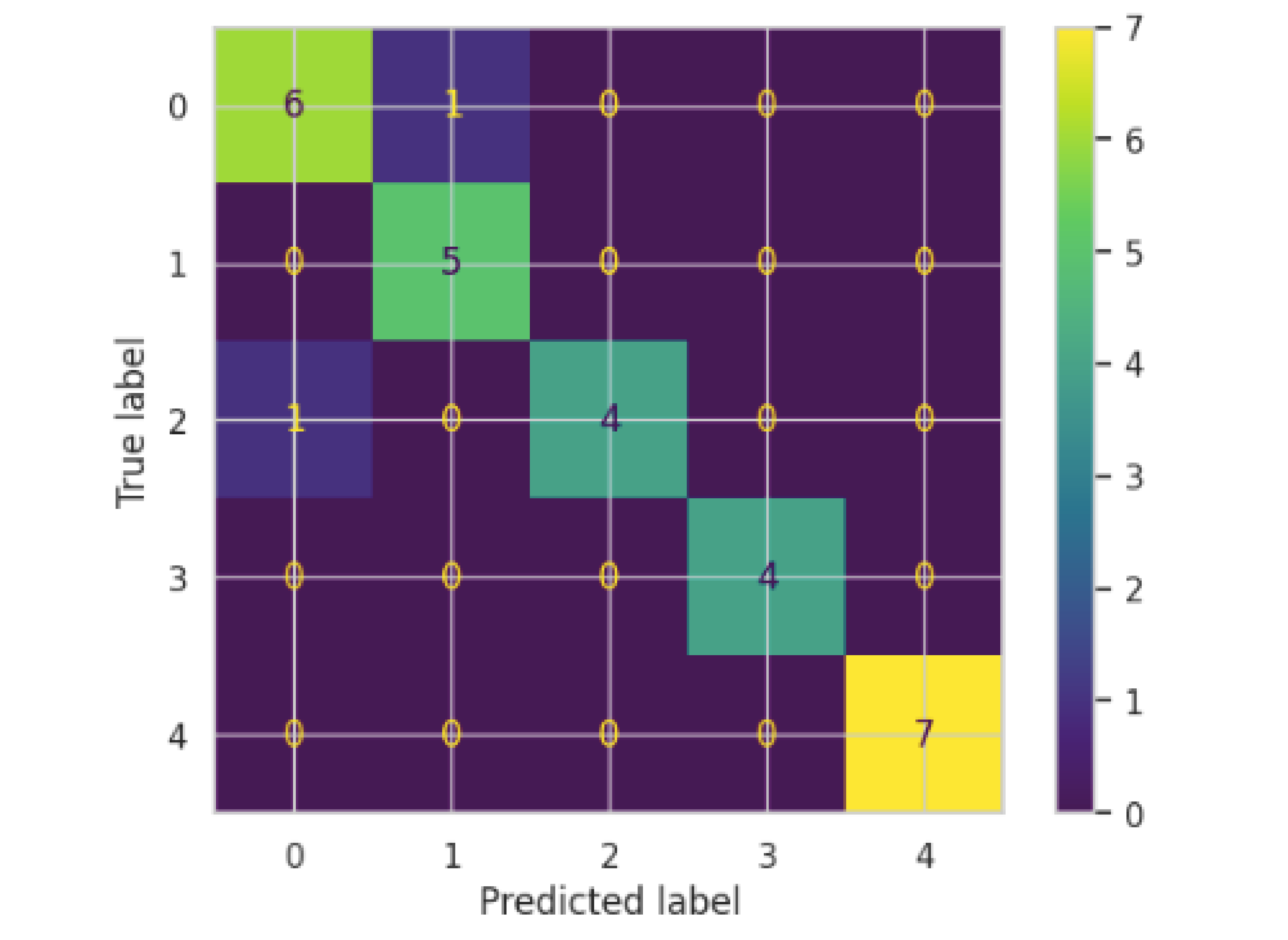
1.Baseline CNN Model

- Achieved high training accuracy (99.17%) but suffered from overfitting(validation accuracy only 82.14%).
- Struggled with **generalization**, leading to high test loss and misclassifications.

2.Improved CNN Model (Regularization & Augmentation)

- Enhanced CNN with Dropout, L2 Regularization, and Data Augmentation to reduce overfitting.
- Improved validation accuracy (85.71%) but test loss remained relatively high.
- Better recall across classes but still had some misclassifications.

Confusion Matrix



- Class 0 (Embassava): 6 correct predictions, with 1 misclassified as Class 2 (Super Metro).
- Class 1 (KBS Shuttle): 5 correct predictions, with no misclassifications.
- Class 2 (Super Metro): 5 correct predictions, with no misclassifications.
- Class 3 (City Hopper): 3 correct predictions, with 1 misclassified as Class 4 (City Shuttle).
- Class 4 (City Shuttle): 7 correct predictions, with no misclassifications.

Objective 3: Bus fleet image classification

3.Modified MobileNetV2 (Fine-Tuned & Optimized)

- Model with fine-tuned MobileNetV2 layers, SGD optimizer, and regularization.
- Achieved the highest test accuracy (86.21%) with the lowest test loss (0.4115)
- Most balanced confusion matrix, meaning fewer misclassifications and better overall predictions.

4.EfficientNetV2 (Fine-Tuned & Optimized) - Best Performing Model

- Training accuracy reached 100% from about epoch 13 onward.
- Validation accuracy stabilized at around 92.9% over the later epochs, and the test evaluation reported nearly 89.7% accuracy.

Challenges:

- Data Quality Issues – Images had inconsistent resolutions, noise, artifacts, and lighting variations
- Manual labelling and annotation - Time-consuming, prone to errors, and may include ambiguous classifications.
- Object Detection Difficulties – Occlusions, scale variations, and visually similar objects pose challenges..

Conclusions:

- PSV Classification: Transfer learning (MobileNetV2) improved validation accuracy to 97-98%, making the model reliable.
- Overloading Detection: Fine-tuned MobileNetV2 achieved 94% accuracy with balanced precision and recall.
- Data Quality: Cleaning, augmentation, and standardization enhanced model robustness.
- The original YOLOv8m model was selected due to its better balance of recall and accuracy for object detection.
- Real-Time Efficiency: The original YOLOv8m model is faster (9.3ms vs. 10.2ms), which is important for real-time vehicle tracking.
- Bus fleet image classification EfficientNetV2 model design and training strategy minimized overfitting, leading to superior generalization

Recommendations:

- Model Enhancements: Use ensemble models or self-supervised learning for better classification.
- Data Expansion: Collect more real-world data and generate synthetic samples.
- Deployment: Implement real-time systems with edge computing for traffic monitoring.
- Law Enforcement Integration: Develop a mobile app for on-the-spot vehicle classification.
- Seek High-Quality Video Data: The search for reliable video data has been challenging. To improve the model’s performance, it's essential to obtain higher-quality footage.
- Prioritize YOLOv8m for Real-Time Deployment: Given its balance of speed, accuracy, and recall, the original YOLOv8m model should be used for real-time traffic monitoring systems

Deployment

THANK YOU

ROAD
ENDS

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